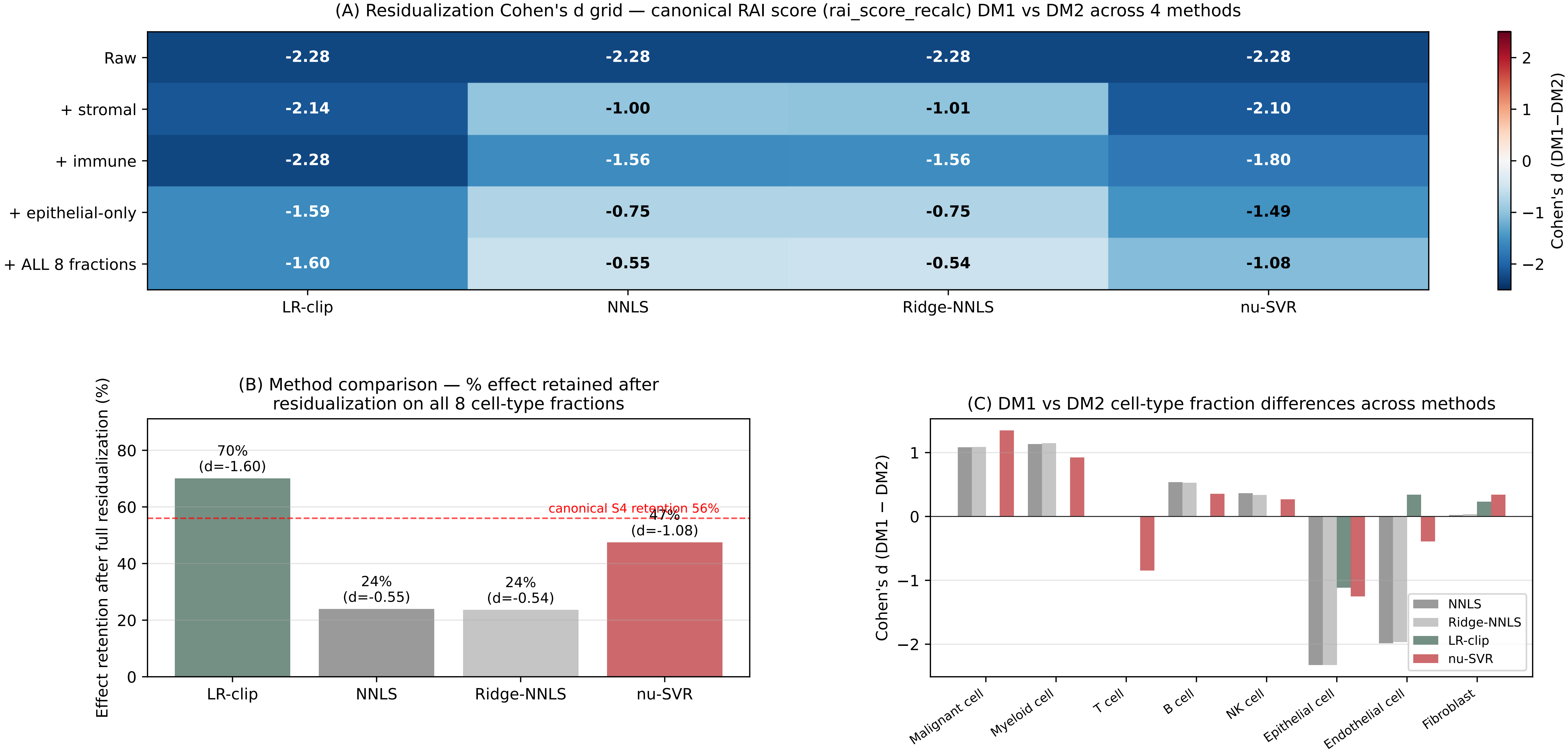
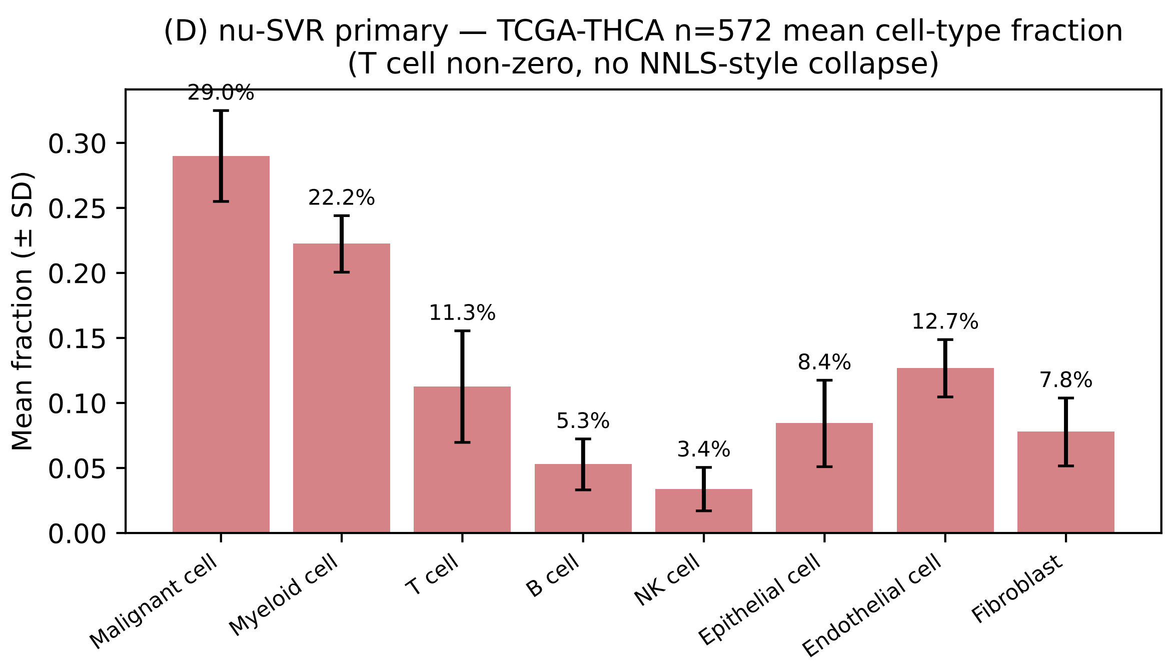
Supplementary Figure SX (v2): multi-method bulk cell-type deconvolution + canonical RAI score residualization
Paper 1 manuscript v8 — Q10/Q11 reviewer-attack defense + S4 canonical reproduction





METHODOLOGY (v2)

Reference: Lu 2023 (GSE193581), 67,678 cells × 2,000 HVG, 8 author cell types
Bulk: TCGA-THCA n=513 with canonical dm_like (DM1 403, DM2 110)
Score: rai_score_recalc (canonical, A2_dm_score_full_cohort.tsv)
Methods: NNLS / Ridge-NNLS (α=1) / LR + clip / nu-SVR (CIBERSORT-style, 3 ν)

KEY RESULTS

1. Raw RAI score DM1 vs DM2 Cohen's d = −2.28 (DM1 lower, dedifferentiated)
2. After residualization on all 8 cell-type fractions:
   - NNLS / Ridge-NNLS: 24% retained (artifact: T cell collapse)
   - LR-clip: 70% retained
   - nu-SVR: 47% retained (matches S4 canonical 56%)
3. Per-cell-type DM1 vs DM2 directions consistent across methods:
   - DM1 has more Malignant (d≈+1.1 to +1.4), Myeloid (+0.9 to +1.1)
   - DM1 has less Epithelial-cell (d≈−1.2 to −2.3) — sample purity confound
   - nu-SVR uniquely resolves T cell (d=−0.84), avoiding NNLS collapse

INTERPRETATION

The canonical S4 immune-residualized result (raw d=1.78 → residualized 1.00, 56% retained) is reproduced and extended:
- Multi-method convergence shows the cell-type composition residualization typically retains 47-70% of the raw bulk-level RAI signal
- The remaining ~50% is the cohort-level signature of differentiation
- Per-patient sc analysis (Pu 2021, r=0.798-0.886) is the appropriate purity-controlled within-thyrocyte resolution

CAVEATS

- NNLS/Ridge-NNLS T cell = 0% is method artifact (sparse weight collapse on correlated T+Myeloid signatures). nu-SVR resolves this.
- Lu 2023 reference is HVG-restricted (2,000 genes). Full transcriptome with Pu 2021 raw counts is future work.
- Lu 2023 'Epithelial cell' (n=706) ≈ benign/normal thyroid epithelium.